Royal Devon University Healthcare NHS Foundation Trust

Physiotherapy Guideline for: **Management of Haemoptysis in Cystic Fibrosis – Trustwide**

SUMMARY

This guideline supports Physiotherapists working in the specialty of Cystic Fibrosis in the management of patients with haemoptysis

KEY POINTS

- It gives guidance on how a physiotherapist should respond to an episode of haemoptysis in a patient with Cystic Fibrosis
- It defines haemoptysis according to volume which enables a physiotherapist to follow the correct line of treatment

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to Physiotherapy staff whilst leaving room for professional judgement and adaptation, to fit individual circumstances.

1.2 Haemoptysis can range from a minor streaking of blood in the sputum to massive bleeding that can have life-threatening consequences due to airway obstruction, hypoxaemia and haemodynamic instability. Haemoptysis can occur in up to 60% of patients with cystic fibrosis (CF) and becomes more frequent as their disease progresses. It can be caused by persistent inflammation of the airways and vascular growth which results in hypertrophied bronchial arteries. Chronic and acute inflammation weakens the vessel walls and often leads to bleeding into the bronchial lumen.

1.3 Admission can be unplanned and often out of hours. If a patient presents in North Devon District Hospital (NDDH) they should be discussed with the respiratory or paediatric consultant on call at the RD&E hospital via switchboard on 01392 411611 out of hours or during normal working hours direct with the CF team via epic messaging or 01392 402682 (paeds), 01392 402726 (adults). In the case of massive haemoptysis, the patient will need to be managed acutely involving HDU/ WATCH (retrieval) teams as needed alongside the Exeter CF centre. If scant, mild or moderate haemoptysis, the guidance for physiotherapy management can be followed where they present

1.4 This can be a frightening situation for the patient; especially if it is a first episode. There may be an inclination of the patient to avoid airways clearance and in some cases the on-call physiotherapist will be asked to assess and offer advice in how to manage airway clearance appropriately.

2. BACKGROUND

2.1 This is a revision of an existing clinical guideline that is due for renewal.

3. DEFINITIONS

3.1 *Haemoptysis – the expectoration of blood from the lower airways*
Embolisation – a procedure that uses tiny particles to block a blood vessel
ACT – airway clearance technique
AD – autogenic drainage
ACBT – active cycle of breathing techniques

HDU – High dependency Unit
WATCh – Wales and West Acute Transport for Children

4. MAIN BODY OF GUIDELINE

4.1 When should the on-call physiotherapist see the patient?

- The CF team will aim to see elective and emergency admissions made during the working day, Monday to Friday.
- Admissions outside the working day should be referred to the **on-call physiotherapist** or weekend physiotherapy team through switchboard after discussion with the medical team.

4.2 How should the patient be treated?

- It is likely that the CF Consultant/ CF Physiotherapist will have made a written medical plan in the patients notes.
- If this is a new episode or out of normal working hours, assess the patient and identify the current clinical need, looking for evidence of sputum retention and breathlessness.
- Treatment depends on the severity of the haemoptysis. Estimating the volume of blood expectorated is challenging. Please document the volume of blood the patient has expectorated. Encourage the patient to expectorate further sputum/blood into a disposable sputum pot to aid monitoring.
- The volume of blood expectorated is measured over a 24 hour period, therefore treat cautiously if they are within the first 24hrs from starting bleeding.
  - **Scant haemoptysis: <5mls over 24 hr period**
  - **Mild haemoptysis: 5-50mls over 24 hr period**
  - **Moderate haemoptysis: 50-250ml over 24 hr period**
  - **Massive haemoptysis: >250ml over 24 hr peri**

- If **mild haemoptysis:** continue airway clearance, inhaled therapies and exercise, but with close monitoring of symptoms combined with medical management to treat underlying probable infection. Care is taken to suppress cough and to advise gentle ACT using AD or ACBT avoiding increasing intra thoracic pressures if possible.

- If **moderate haemoptysis:** physiotherapy interventions should be modified following clinical assessment. This should include careful consideration of the benefits against the risks of altering treatments regimes, recognising that maintaining effective airway clearance is an important component of ongoing

treatment of underlying infection. Treatment modifications should involve the following:

- Avoiding positive pressure techniques and manual techniques (except gentle overpressures) during, and for 24-48 hours post bleed.
- Consider ACT such as ACBT or AD which utilise controlled coughing.
- **The use of hypertonic saline (HTS) and Dornase Alfa (Pulmozyme/DNase) nebulisers** have been suggested to present a risk, although this remains unproven. Consider stopping these on an individual basis if they are identified as a potential trigger for uncontrolled coughing or further bleeding. HTS might be considered more of a risk due to its potential to irritate the airways, induce cough and therefore possibly provoke bleeding. Nebulised 0.9% saline may be beneficial as an alternative to hypertonic saline if difficulty clearing thick secretions
- Temporarily ceasing Non-invasive ventilation (NIV), or, where this is considered too great a risk, consider temporarily reducing inspiratory pressures until bleeding settles.

- In **massive haemoptysis:** the aim of initial management is maintenance of gas exchange by maintaining the airway, administering oxygen, and careful positioning. Nebulised adrenalin may be administered where frank haemoptysis does not settle spontaneously.
    - Temporarily cease all physiotherapy treatments and exercise. Minimise coughing and aim to position the patient in high side lying with their bleeding side (if known) down.
    - Optimise oxygenation and humidification – confirm oxygen prescription
    - Bronchoscopy, the prescription of tranexamic acid or CT scan +/- embolisation maybe indicated. Following embolisation, in liaison with interventional radiologist, resume normal airway clearance and a graduated exercise programme. Ensure adequate analgesia if chest pain limits effective airway clearance.

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, complaints or claims will be reviewed as per Trust policy and may result in an audit and / or amendment to the guideline.

Monitoring that this guideline is being followed will take place through the existing training programme in place for all on-call Physiotherapists

6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

Standards of Care and Good Clinical Practice for the Physiotherapy Management of Cystic Fibrosis (2020)

Incident Reporting, Analysing, Investigating and Learning Policy
Complaints Policy and Procedure

7. PUBLICATION DETAILS

Author of Clinical Guideline	*Alice Day, Physiotherapy Clinical Lead in paediatric CF*
Division/ Department responsible for Clinical Guideline	*Clinical Specialist Services/Physiotherapy*
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